MIL-C-24643/27B
3 October 1986
SUPERSEDING
MIL-C-0024643/27A(SH)
8 November 1985
MIL-C-24643/27
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LS2AU (INCLUDING VARIATIONS LS2A AND LS2AUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be supplied in three variations: LS2AU (unarmored), LS2A (armored) and LS2AUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 22-7. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.024 inch. Telephone identification code applied by method 3.
Fourth - Two conductors cabled together to form a pair, with nominal lay of 1-1/2 inches.
Fifth - Forty pairs cabled together, with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.
Sixth - Binder tape, applied helically with overlap.
Seventh - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS2AU

Eighth - Binder tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. (Nominal thickness 0.060 inch). Cable surface marking required.

Construction variant for type LS2A

Eighth - Binder tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Nominal thickness 0.060 inch).
Tenth - Braided metal armor and paint.

Construction variant for type LS2AUS

Eighth - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25-percent minimum overlap.
Ninth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. Nominal thickness 0.060 inch. Cable surface marking required.

TABLE I. Details.

Military 1/ part no. M24643/27	Size 2/	Number of pairs	Overall diameter (inches)					
			Type LS2AU		Type LS2A		Type LS2AUS	
			minimum	maximum	minimum	maximum	minimum	maximum
-01U0	40	40	1.320	1.370	1.370	1.420	1.380	1.430

1/ Unarmored, overall shield. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LS2AU, LS2A and LS2AUS.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield.....	2500
Shield to water.....	1000
Insulation resistance - conductor to conductor megohms/1000 feet, minimum.....	5000

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EXAMINATION AND TESTS: (Continued)

	Requirements
Shield to shield (type LS2AUS only, overall shields only), volts, minimum	6.0
Megohms/1000 feet, minimum	1.0
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance - at 1 MHz, pF/ft maximum	
Mutual capacitance	20
Group B:	
Cold bending, cable - at 0 ± 2°C, mandrel diameter 12 times cable diameter	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum	25
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing jacket (types LS2AU and LS2AUS only) cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Shield conformance to material, construction and coverage	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Armor (type LS2A only) conformance to material, construction and coverage	No failure
(B) Surface transfer impedance (types LS2AU and LS2AUS only)	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-27)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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